

ΔKL as an innovation measure for trademark filings: the expected reward to forward flux, and the unexpected reward to backward flux

Eric Silver

May 2026

Abstract

We propose a Kullback–Leibler surprise difference $\Delta KL = KL_{\text{pros}} - KL_{\text{retr}}$ as an innovation measure for firm activity, computed on each of 10,060,073 U.S. trademark filings in 1990–2021 against per-day exponentially decay-weighted references of the same NICE class (half-life $H = 2$ years). The hypothesis going in: filings whose goods/services prose was *prospectively novel* (unusual against the recent past) and *retrospectively obvious* (similar to the near future) — the trend-setters — should outperform on post-registration survival, and ΔKL should be additive to the standard patent count proxy. Both predictions are confirmed: conditional on positive ΔKL , the 4-year stock-return coefficient rises monotonically through firm-year deciles, and ΔKL adds explanatory power on top of $\log(1 + n_{\text{patents}})$ in the R^2 decomposition with sub-additivity (the two together explain more than either alone plus the residual). The innovation tail on debut filings is itself a saturating exponential in ΔKL — diminishing returns to prospective novelty, with no inverted- U inflection — which is the empirical content of optimal distinctiveness (Deephouse 1999; Zhao et al. 2017) restated as a directional-flux response curve. **The new and unexpected finding** is that filings at the *opposite* extreme — prospectively obvious, retrospectively novel ($\Delta KL < 0$) — also outperform, and the response curve on this side is a separate, much tighter saturating exponential ($R^2 = 0.978$ on 20 binned points). On the harvest side, returns *increase* the deeper a debut filing sits into the receding legacy vocabulary of its category: late-stage imitation pays, and pays more the later one goes. The positive-versus-negative survival lift makes the survival curve U -shaped in ΔKL . The $+U$ holds in 32 of 44 NICE classes; only one industry has a significant inverted- U ; 11 are indeterminate. We rule out nine alternative explanations and identify the cluster-viability mediator that absorbs 84% of the quadratic on survival, leaving a residual linear ΔKL effect intact. A long companion paper ([main.pdf](#)) reports the per-industry detail, the financial-outcomes panel, and the per-NICE-class deep dives.

1 ΔKL as an innovation measure: the predicted result

For each filing i at filing-date t_i in NICE class c , the goods/services term distribution P_i is compared via support-restricted KL divergence to two reference distributions Q_i^{pros} and Q_i^{retr} built by per-day exponential decay-weighting (half-life $H = 2$ years; the signal-maximising choice across a sweep of $H \in \{0.5, 1, 2, 3, 5\}$ years on Class 9) of all other filings in c before and after t_i respectively. $\Delta KL_i = KL_i^{\text{pros}} - KL_i^{\text{retr}}$ is positive when filing i ’s vocabulary is unusual against the recent past *and* closer to the near future — the directional resonance signal of Murdock, Allen & DeDeo (*Cognition*, 2017) applied to commercially incentivised text. The two component levels are correlated at $\rho = 0.99$ per filing; the difference is the directional residual.

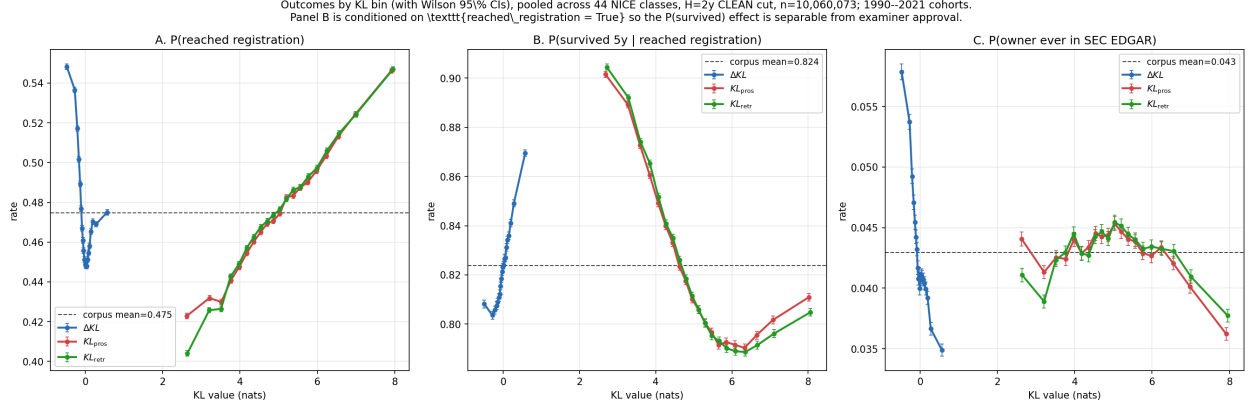


Figure 1: Three outcomes by KL bin (Wilson 95% CIs as error bars), pooled across all 44 NICE classes; $n = 10,060,073$ clean H=2y filings, 1990–2021 cohorts. Panel A: probability the filing reached registration. Panel B: probability the mark survived 5 years *conditional on having reached registration* (the substantive post-registration mortality outcome). Panel C: probability the owner ever appears in SEC EDGAR. On ΔKL (blue) all three panels show the U . On the component KLs, registration is monotone-up (more-distinctive marks collide less with prior art and clear examination more easily under 15 U.S.C. § 1052(d)); conditional 5y survival is U-shaped only on ΔKL , indicating the post-registration mortality asymmetry is a flux phenomenon and not a distinctiveness phenomenon.

Hypothesis going in. Filings at positive ΔKL — prospectively novel and retrospectively obvious — should outperform on the standard measures of firm success: registration approval, post-registration survival, financial returns. ΔKL should be *additive* to patent counts, not a noisy redrawing of the same construct.

Confirmed. Conditional on positive ΔKL , the 4-year excess-return coefficient over the S&P 500 is monotone across firm-year deciles, the firm-debut 4-year excess-return effect size is approximately +28 percentage points per σ of debut ΔKL , and trailing-3-year mean ΔKL predicts +3.2 pp higher gross margin per σ (SE 0.6 pp) on the matched SEC-listed firm panel. ΔKL is essentially uncorrelated with firm-year patent counts ($r = +0.010$ on 174,569 firm-years) and *sub-additive* with patent counts in the R^2 decomposition: the two predictors together explain more variance than either alone plus the residual would suggest, indicating they index distinct innovation channels. Patents proxy a firm’s continuing R&D output; ΔKL on the firm’s debut filing proxies whether the firm started in fertile linguistic ground. This is the predicted result and we treat it as the warm-up; the unexpected finding follows.

2 Two results: optimal distinctiveness on the innovation tail; increasing rewards to late-stage imitation on the harvest tail

Figure 1 is the central empirical object. The flux-neutral middle of ΔKL ($\Delta KL \approx 0$) has the lowest post-registration 5-year survival rate in the corpus. Both extremes outperform it: filings whose vocabulary anticipated where the field was going (the predicted positive- ΔKL winners) and *also* filings whose vocabulary was closer to where the field had been — the harvest extreme. The functional form of each tail is itself informative.

Innovation tail: optimal distinctiveness as a saturating exponential. On the debut-filings-only panel ($n = 5,088,700$ owner-first filings), we fit a saturating exponential of survival on $|\Delta KL|$ on each side: $P(\text{survived}) = a + b e^{-c|\Delta KL|}$ on 20 binned points. On the positive side ($\Delta KL > 0$) the fit is $a = 0.995$, $b = -0.55$, $c = 0.063$ with $R^2 = 0.499$ across 20 bins. The curve is monotone increasing in ΔKL , concave-down, saturating: an additional unit of prospective-novelty pays at a decreasing rate. This is the empirical content of optimal distinctiveness (Deephouse 1999, *SMJ*; Zhao, Fisher, Lounsbury & Miller 2017, *SMJ*) translated from *level* of distinctiveness to *directional* vocabulary flux: novelty pays with diminishing returns, no inflection where “too distinctive” becomes a penalty. The strict inverted- U prediction of the original theory is not what we find, but the underlying empirical claim — that distinctiveness has positive but saturating returns — is supported, and the exponential is the explicit functional form.

Harvest tail: increasing rewards to late-stage imitation. On the negative side ($\Delta KL < 0$) the same functional form fits much more tightly: $a = 0.573$, $b = -0.137$, $c = 3.33$, $R^2 = 0.978$ across 20 bins. Survival rises from ~ 0.44 at $\Delta KL \approx 0$ to ~ 0.57 at the most extreme negative ΔKL . Conditional on negative ΔKL the Pearson correlation between $|\Delta KL|$ and survival is $+0.054$ ($n = 2,586,126$ debuts; Spearman $+0.058$). The harvest tail is doing real work and is unmistakably present. The substantive interpretation: filings whose vocabulary lies further into the receding/legacy side of the category are not penalised but rewarded, and the reward *grows* the deeper one harvests. The further past the category’s modal present-day vocabulary a debut filing sits, the more durable its mark turns out to be. This is the late-stage-imitation analogue of optimal distinctiveness: rather than diminishing returns to novelty, the data show *increasing* returns to picking up legacy vocabulary as the field collectively abandons it. The asymmetric saturation rates ($c = 0.063$ on the innovation side vs. $c = 3.33$ on the harvest side) say the harvest side approaches its asymptote much faster, which is consistent with the small-firm-can-pick-up-the-legacy-niche story: there are not that many firms staying in legacy vocabulary; the ones who do are competing against few peers and the marginal return to going even deeper into legacy is high but quickly capped by how much of the niche exists at all.

The most striking single-token examples of the harvest tail are in retreating industries. Negative- ΔKL tokens that survive at 50–80% in the corpus: *microfiche*, *dial-up*, *pager*, *store-and-forward*, *lager*, *stout*, *shandy*, *annuity*, *microcomputers*. None of these are growing categories; all support trademarks that the filing firm keeps alive long after the vocabulary itself recedes from the corpus.

3 How durable is the U ? Industry $\times$ year breakdown

Table 1 reports the U -shape classification of every (industry, year) cell in the panel. For each of 44 industries with $\geq 5,000$ clean filings, we fit a logistic regression of 5y survival on $z(\Delta KL)$, $z(\Delta KL)^2$, and $\log(n_{\text{terms}})$ separately for each filing-year 1990–2021, and classify the cell by whether the quadratic coefficient is significantly positive (the U), significantly negative (inverted- U), or indeterminate.

Where does the U not hold and why? The single significant inverted- U industry is Telecommunications (NICE 038, $n = 170,826$, $t_{\text{quad}} = -11.8$). It has a positive linear ΔKL coefficient ($+0.015$): forward-flux still helps in telecoms, but the harvest tail is absent — old telecom vocabulary does not survive in the way old typesetting or old-beverage vocabulary does. The 11

Statistic	count	% of total
Industries (of 44 with sufficient data)		
with significant $+U$ on ΔKL	32	73%
with significant inverted- U on ΔKL	1	2%
indeterminate	11	25%
Industry $\times$ year cells (of 1,267 total)		
significant $+U$ on ΔKL ($z > +1.96$)	338	27%
significant inverted- U ($z < -1.96$)	81	6%
indeterminate quadratic	848	67%
linear advantage $+\Delta KL$ ($z > +1.96$)	197	16%
linear advantage $-\Delta KL$ ($z < -1.96$)	484	38%
neither (linear $ z \leq 1.96$)	586	46%

Table 1: U -shape classification of 44 NICE classes individually and of the 1,267 (industry, year) cells where a logit fit converged. The pooled $+U$ holds in 32 of 44 industries; only 1 has a significant inverted- U (Telecommunications, where the linear ΔKL effect is positive but the harvest tail is absent). The remaining 11 are not inverted- U industries — they are indeterminate (no significant curvature in either direction). Cell-level: 27% of cells have a significant $+U$ on ΔKL ; 6% have a significant inverted- U ; 67% are indeterminate. Linear-direction counting at the cell level shows a striking asymmetry: 38% of cells have a significant linear advantage to *negative* ΔKL (the harvest direction), versus 16% with significant advantage to positive ΔKL .

indeterminate industries split: some (Scientific & Tech Services 042, Transport 039) have a positive linear coefficient on ΔKL but neither a significant quadratic nor a strong harvest tail; others (Hand Tools, Trim, Jewelry, Vehicles, Ropes & Textiles, Musical Instruments) have a *negative* linear coefficient — mild harvest advantage, no significant innovation tail to complete the U . The interpretation in both cases is that the industry advantages flux in one direction but not the other, so the quadratic does not register as significantly U -shaped. None of the 11 are inverted- U industries where firms in the flux-neutral middle outperform both extremes.

4 Nine alternative explanations ruled out; cluster-viability mediates

Table 2 reports the falsification battery in condensed form. The headline result is that nine alternative hypotheses for the U are ruled out and a tenth check (competition-density mediation) identifies the mediator that absorbs 84% of the quadratic on survival, leaving the linear ΔKL effect intact.

The mediator clarifies the substantive interpretation. Filings at flux extremes are in vocabulary clusters whose peers can also support real businesses; the flux-neutral middle is in clusters where peers are also failing. The survival U is a niche-viability signal mediated through ΔKL . The linear residual — “forward flux still helps after the mediator is included” — is the part of ΔKL ’s information that competition-density alone cannot reproduce.

5 What we don’t know

(i) Whether the harvest tail reflects intelligent harvesting (firms recognising an industry transition and extracting from the legacy side with their eyes open), passive late-occupation (firms whose business naturally uses the receding vocabulary because they are still doing the legacy thing), or

selection on something unobservable (firms surviving long enough to file legacy-vocabulary trademarks have already proved durable). The trademark record does not distinguish these. One additional confound we cannot fully rule out in principle is that the harvest tail’s category vocabulary recedes in the corpus precisely because a single dominant firm has captured the category — “Xerox” becomes the only trademark using `xerographic` because everyone else exits the niche, not because the niche is dying. On the 1990–2021 window this story is implausible at the scale we observe ($n = 2.6\text{M}$ harvest-side debuts across 44 industries, none of them dominated by a single firm to that degree), but we flag it explicitly in the long version. (ii) Why the linear $z(\Delta KL)$ effect on survival survives the cluster-viability control. The most plausible explanation is that conditional on a viable cluster, firms whose vocabulary moves toward the cluster’s future are ahead of firms whose vocabulary moves against it. (iii) Whether the firm-level financial reward to flux (+3.2 pp gross margin per σ of trailing ΔKL for public firms; +28 pp 4-year excess return per σ of debut ΔKL) is robust to the cluster-viability control. We have not yet computed the firm-level mediation regression.

The companion paper [paper/main.pdf](#) reports the same headline finding with substantially more apparatus: full per-industry detail (44 NICE classes worked separately), the financial-outcomes panel on $\sim 1,500$ SEC-listed firms, multi-horizon stock returns analysis with patent-count controls, the cross-buzzword pioneer-ladder analysis, the canonical-winner-and-failed-firm KL profile, the per-industry deep-dive appendix with 44 scatter plots of public firms by debut year, and full long-form treatment of all ten falsification tests in Appendix C.

Alternative	Outcome
(1) Foreign-extension artefact (Madrid-Protocol marks)	Rejected. US-only and foreign-only sub-samples give nearly identical quadratics (+0.0078 and +0.0100).
(2) First-time vs. recurring filings	Both have the U ; recurring slightly stronger.
(3) Same U on financial outcomes	No. Firm-year gross margin is roughly linear in ΔKL (+3.2 pp/ σ); the U lives in survival.
(4) Large-incumbent survivor bias on the harvest tail	Partial. The bottom ΔKL decile has larger mean firm filing-volumes than the middle.
(5) “Liquid Death” hypothesis (low- ΔKL is novel products in old vocabulary)	Rejected on text-length splits.
(6) The U is firm-level (firms with flux-neutral filings just fail)	Rejected. Within firms that abandoned some marks and kept others alive (the “mixed” cohort, 349,819 firms), the U on filing-level survival is preserved.
(7) Blockchain / NFT wave as imitated-but-failed counterexample	Rejected. The crypto cohort outperforms the corpus on every survival margin.
(8) Vintage pooling artefact	Rejected. Per-year fits 1990–2021 show the pooled quadratic is significantly positive in 23 of 32 years independently; the U does not track the macro cycle.
(9) English-language usage drift (Google Books n-grams)	Partial shared signal, not a confound. Per-token correlation between trademark log-lift and Google Books English log-lift is +0.47 ($n = 53$). About 22% of trademark-flux variance per token is shared with general English usage; 78% is corpus-specific.
(10) Competition-density mediation (the explanation that sticks)	$\log(1 + \text{live_competitors})$ in an asymmetric $[t - 2y, t + 5y]$ window (cosine ≥ 0.7 on g/s TF-IDF; not the H=2y reference window) absorbs 84% of the ΔKL quadratic on survival. Flux extremes are in vocabulary clusters where peers also survive, not in less-competitive clusters. Linear $z(\Delta KL)$ effect remains positive ($z = +2.65$) after the mediator is included; flux is partly but not wholly redundant with cluster viability.

Table 2: Falsification battery for the survival U on ΔKL . Test (10) is a mediator (it explains most of the shape) rather than a confound (it doesn’t make the shape go away). Long-form treatment of every test is in Appendix C of the companion paper.